ShopPilot — Agent-Ready Ecommerce, One Install

Business whitepaper · 27 September 2026 · updated 21 October 2026 · Prepared with live experimental evidence (74% ingestion reduction, 3/3 task accuracy) · Confidential draft for discussion

1. Executive Summary

AI agents are starting to shop on behalf of people — but today they do it blind: reading page layouts, guessing at buttons, breaking on every redesign. **ShopPilot** is a browser extension that, with one install, gives any AI assistant structured tools (`search_products`, `add_to_cart`, `checkout`) on virtually every online store — using each store's own real APIs and the user's logged-in session. No merchant signup. No per-site configuration.

The wedge: ecommerce platforms are standardized under the hood. **4 platform adapters (Shopify, WooCommerce, Magento, BigCommerce) cover an estimated 60%+ of global storefronts.** Our controlled experiment measured a **74% reduction in agent data ingestion at equal accuracy** — the founding evidence for the product thesis.

74%	**4**	**₹0**
less agent data ingestion (measured)	platform adapters → thousands of stores	cost to start (extension + affiliate, self-funding)

2. The Market Shift (Why Now)

- **The standards are real:** WebMCP is incubating at the W3C (authors from Google & Microsoft); Chrome opened an origin trial (v149, Feb 2026); OpenAI's ChatGPT browser already ships a version ("Site tools"). Sites that declare tools get agent traffic first.
- **Agent commerce is arriving with or without merchants:** ChatGPT's Instant Checkout and Operator, Perplexity Shopping, Amazon Rufus — the assistant layer is being built now. What's missing is the *last mile*: letting agents act on the long tail of stores that will never build their own agent integration.
- **Historical precedent:** Honey solved "apply coupons on any store" as an extension with zero merchant cooperation — 17M+ users, acquired for ~$4B. Rakuten/Ebates built a multi-billion business on the same "ride the user's session" model. ShopPilot is the same shape, one layer deeper: not just discounts, but the entire act of shopping.

Example (target user): Priya tells her voice assistant on the drive home: "Order my usual coffee machine descaler and find a birthday gift under ₹2,000 for Meera — something highly rated." The agent, via ShopPilot, searches Amazon India + a Shopify boutique + Flipkart, compares ratings, adds the gift to the cart, and sends Priya one confirmation prompt before checkout. Total interaction: two sentences and one tap. Today this requires opening 4 apps and 20 minutes.

3. Product

3.1 What the user installs

One Chrome/Edge extension. After install: every store the user shops at exposes the same 5 structured tools to their AI assistant. The extension detects the platform (Shopify/Woo/Magento/BigCommerce), injects the tool layer wired to that platform's standard endpoints, and runs inside the user's session — no credentials stored, no data exfiltrated. A generic form-annotation fallback covers long-tail platforms.

3.2 Who pays, at a glance

Model	Who pays	When	Why it works
A. Affiliate commission (launch)	Merchant (per order)	Every agent-driven checkout carries affiliate attribution (1–8%)	Clean attribution: the agent's click *is* the referral. Self-funds the free product from day 1
B. Pro subscription (month 3+)	Consumer	₹150–300/mo: unlimited stores, checkout automation, price-drop alerts, cross-store comparison	Pays only when the product demonstrably saves money/time
C. Merchant SaaS (month 6+, the moat)	Merchant (subscription)	$29–99/mo to be in the agent registry + analytics dashboard + placement in agent feed	"SEO for the agent era" — merchants pay for agent visibility before agents become a traffic channel they can't measure
D. SDK licensing (year 2)	Platforms/SIs	License adapters to voice-agent & ecommerce SaaS vendors	They need platform coverage; we own the adapter library

Deliberately excluded: transaction/payment intermediation (RBI/PCI burden, not core to value), selling browsing data (kills trust, Chrome policy risk).

> **Example (Model C pitch to a merchant):** "Right now, when an AI agent visits your store, it sees an unstructured wall of HTML — it may mis-price, mis-order, or skip you entirely, and you'll never know. For $49/month, ShopPilot registers your catalog in the agent tool registry, and you get a dashboard: 312 agent visits this month, 41 carts, 19 completed orders worth ₹86,000 — all attributed."

4. Evidence: The Founding Experiment

Before any code of ShopPilot, we built two functionally identical stores (demonomcp.converso.work vs demowithmcp.converso.work) differing *only* in whether agents get structured tools or the raw page, then ran identical shopping tasks with an instrumented agent driver. Full technical report available; headline results:

Metric (3 shopping tasks)	Plain site	Tool-enabled	Delta
Data the agent must read	33,971 B	9,157 B	−74%
LLM input tokens (≈4 B/token)	~8,500	~2,300	−73%
Task success	3/3	3/3	equal
Product selection	identical picks		equal

> **Independent verification (Oct 2026):** the live demo's public tool manifest was independently read and fingerprinted by **Dominion Observatory** (an MCP-manifest monitoring firm) on 18 October 2026 — 5 tools, SHA-

256 `d603406d…70b57` — reproducing *exactly* against our published baseline (github.com/kamatanuj/webmcp, `mcp_baseline.json`). CI on every push now fails on any silent change to that manifest, so agents always build against a verified, unchanged contract.

The gap scales with catalog size: the plain-path agent always reads everything; the tool path reads only what the server filters to. **On a real 10,000-SKU store, the delta isn't 74% — it's ~99%.**

Example (why tokens = money): an agent browsing a plain Flipkart-scale catalog reads ~8,500 tokens per task set; with tools, ~2,300. At typical LLM pricing that's a ~3.7× cost difference per task set — multiplied by every search, every comparison, every retry. For voice agents (real-time latency budgets), it's the difference between a 6-second and a 1-second response.

5. Market Sizing (directional)

- **TAM:** Global affiliate marketing spend ≈ $15–17B/yr (2025–26 est.); ecommerce conversion-tool SaaS (Honey-class, price engines, cart recovery) ≈ $8–12B. Agent-commerce is a redistribution inside these pools.
- **Beachhead SAM:** ~5.5M+ Shopify stores, ~700k+ WooCommerce-active, plus Magento/BigCommerce — India angle: D2C boom (100k+ Shopify India stores, UPI checkout flows standard). If 1% of Shopify stores buy merchant SaaS at $49/mo ≈ **$26M ARR from one platform alone**.
- **Wedge SOM (year 1):** 50k extension installs (0.05% of Honey's base), 2% Pro conversion (500 × ₹200/mo ≈ ₹12L ARR) + affiliate on ~₹4–8L/mo GMV through agent checkouts + 50 merchant SaaS pilots (₹20–45L ARR).

6. Competition & Differentiation

Player	What they do	Why ShopPilot wins
WebMCP Hub, Customaise (extensions)	Generic per-site tool injection via community configs	We cover stores by *platform*, not per-site: one Shopify adapter = every Shopify store, instantly. Community configs can't match platform-level coverage
ChatGPT Site tools / Chrome WebMCP	Browser-native standard; needs sites to adopt	We are adoption-agnostic: work in any browser, on stores that never adopt. Complementary — we can speak the native API when it lands
Honey/PayPal, Cuelinks, EarnKaro	Coupons/affiliates on user's session	They monetize the session but don't *drive* it. We're the actuation layer they'll need when users shop by voice/agent
Amazon Rufus / Flipkart assistant	On-platform, closed	We're cross-platform by design — assistants don't have store loyalties, users do

7. Go-to-Market

- **Phase 1 (weeks 1–8):** MVP extension + Shopify adapter + affiliate layer. Launch on Product Hunt, r/Shopify, D2C India communities. Target metric: agents successfully completing a real checkout on a real store, end-to-end.
- **Phase 2 (months 3–6):** Pro tier + price tracking; WooCommerce adapter. Growth loop: every checkout confirmation dialog is a share moment ("your AI shopped this — install ShopPilot").

- **Phase 3 (months 6–12):** Merchant SaaS. Convert affiliate merchants to registry subscribers ("you're getting orders already — want analytics + placement?"). Shopify App Store listing as distribution.
- **India wedge:** voice-first shoppers (vernacular) + UPI + D2C explosion. A Hindi voice agent that can shop on any store, correctly, is a demo that sells itself.

8. Financial Sketch (Year 1, conservative)

Line	Pessimistic	Base	Optimistic
Extension installs (Dec-27)	10k	50k	150k
Pro subs @ ₹200/mo	150	500	1,500
Pro revenue (₹L)	3.6	12	36
Affiliate (GMV ₹ Cr × avg 3%)	2.4 (₹8 Cr GMV)	9 (₹30 Cr)	27 (₹90 Cr)
Merchant SaaS (pilot→paid)	10 × ₹2k/mo	50 × ₹3k/mo	120 × ₹4k/mo
SaaS revenue (₹L)	2.4	18	57.6
Total ARR (₹L)	**~6**	**~39**	**~120**
Costs (2 eng + infra, ₹L)	30	36	48

Illustrative, not a forecast. Affiliate depends on category mix; SaaS conversion is the key unknown. Bootstrap-friendly: the base case covers costs from month ~9.

9. Risks & Mitigations (MBA hat on)

Risk	Severity	Mitigation
Platform API changes / breaking Shopify endpoints	High	Adapters pinned per platform version; contract tests on real stores; community bug bounty
Chrome Web Store policy (injection into third-party sites)	High	Stay within user-session model (Honey precedent); no credential capture; explicit consent UX; native WebMCP API path when available
Merchant ToS objections	Medium	Registry is opt-in and revenue-positive for merchants; free tier only reads public product data + user's own cart
Payment errors via agent (wrong item, wrong qty)	Medium	Checkout confirm-gated by design (WebMCP spec mandates human-in-the-loop); full audit log of every tool call
Spec drift (WebMCP draft evolving)	Medium	Target the stable polyfill API today; browser-native API is additive — adapters outlive the interface
Big player ships the same thing (OpenAI/Google)	Real	Their agents need third-party stores too — partner path: ShopPilot as their adapter layer. Platform neutrality is the moat; be the Switzerland
Affiliate dependency (programs change terms)	Medium	Diversify via aggregators (Cuelinks/EarnKaro); Pro+Merchant revenue reduces share over time

10. MBA Guidance: What I'd Do (and in what order)

> **Strategic read:** This is a *platform land-grab timed to a standards shift*. The window is 12–24 months before browser-native WebMCP + Big-Tech agents make structured shopping table stakes. Goal isn't to win ecommerce — it's to **own the adapter/registry layer before the standards settle**, then be either the toll booth (merchant registry) or the acquisition (agent platforms buy adapter libraries).

1. **Validate before building (2 weeks, ~₹0):** 15 interviews — 5 merchants (Shopify India D2C), 5 shoppers (voice-assistant users), 5 agent-platform builders. Kill criteria: merchants don't care about agent traffic, or shoppers won't install an extension for shopping. If merchants light up at "agent analytics on my store" → Model C leads the roadmap, not the extension.
2. **Build the thinnest wedge:** extension + Shopify adapter + affiliate tags only. One platform, one revenue line, real stores, real checkouts. Everything else waits for signal.
3. **Instrument obsessively from day 1:** every install, store-visit, tool-call, cart, checkout, conversion — this dataset is the pitch for both Pro and Merchant SaaS. No other player will have agent-behavior data across stores.
4. **Pricing psychology:** free tier must feel magical (one successful agent checkout = the aha), Pro must be priced against visible savings ("saved you ₹1,240 this month" > any feature list), Merchant SaaS must be ROI-framed ("agent orders delivered" not "agent tools exposed").
5. **Structure:** start as a proprietorship/D2C product inside an existing entity for speed; incorporate (US LLC or India Pvt Ltd) before merchant SaaS or any external data touches the product. Affiliate income from day 1 keeps it self-funding — raise only after retention data exists.
6. **The one metric that matters:** monthly agent-driven checkouts per active install. If that number holds above ~1, this is a business; everything else is optics.

> **Bottom line:** the technology risk is low (working demo + known endpoints), the timing risk is low (standards momentum is visible), and the execution risk is concentrated in one honest unknown: *will consumers let an agent finish checkouts?* The affiliate model means you get paid to find out.

Companion artifacts: WebMCP_Comparison_Report.pdf (layman), WebMCP_Technical_Report.pdf (technical), live demos at demonomcp/demowithmcp.converso.work · Contact: Hermes Agent